

M1: Monte Carlo and Nonlinear Spot Checks

9 October 2026 (rerun)

Do finite noisy records behave as the analytic engine predicts, away from p_0 ? Amir Dollie (DLLAMI004), EEE4022S

SHARC Code: `runM1MonteCarloSpotChecks.m`, `m1Tools.m`, `g3EstimatorCore.m`, `resolveM1Scenarios.m`, commit 1b61a0f

Results: `Results/M1/M1_20261009_143730`

Rerun after the amplitude amendment (8 Oct). The EMM FRFs used by the 3 Oct run were low by $\alpha(\omega) = H\omega^2/g$; `computeSensorFRF` now multiplies by α and the twin cache was migrated exactly (`cfg.meta.changes`, 2026-10-08).

Two cases were **re-resolved** by their unchanged pre-registered rules on the rerun source studies (logged 2026-10-09):

- the A1 boundary case, from the MIZ floe to $\beta \times 2$;
- the R2 stress sea, from $\omega_p = 3.5$, $\gamma_J = 7$ to $\omega_p = 6$, $\gamma_J = 7$.

The D1 reference followed the D1 rerun (s18 to s6). Acceptance values are unchanged; tester 12/12. This summary supersedes M1_20261003_221127.

Summary

Question. D1, A1, R1 and R2 rest on *analytic* predictions (E1): the Fisher information F , the standard deviations $\sigma = \sqrt{\text{diag } F^{-1}}$ and the Welch-expected bias $\delta\theta_W$ of one record, without simulating records. Do real finite records follow these predictions in the four deliberately awkward cases fixed by the pre-registered rules? Truth and inverse share the EMM, so M1 verifies the estimator and the analytic approximation, not the physics (V1).

Outcome. Three cases pass every criterion. The R2 case **fails formally on one criterion**: one of its two nonlinear inversions ended flagged “not converged”.

- **Scatter:** empirical over predicted lies between 0.91 and 1.11 in all four cases (band 0.85 to 1.20).
- **Coverage:** the 95% coverage of the predicted ellipse lies between 0.915 and 0.950 (band 0.88 to 0.99).
- **Bias:** every mean offset is consistent with the Welch-expected bias (within 2.0 standard errors).
- **Nonlinear:** all eight inversions end at the linearised estimate (gap below 0.0005σ), starting 290 to 500 σ away. The failing run stopped at the same point as the other start, but through the solver’s “no further descent” exit, which G3’s code does not label as converged.

The analytic predictions hold. The failure is in the termination labelling, and is reported as a failure under the registered rule.

1 What the study does

Why M1 exists. The analytic E1 engine replaced repeated Monte Carlo for D1, A1, R1 and R2. G3 validated the analytic route near p_0 , but those studies deliberately moved into harder parts of the design, forcing and floe domains. M1 tests whether the E1 predictions of Fisher uncertainty and Welch bias still describe real finite-record estimates in four pre-registered stress cases, each probing a different way the approximation could fail.

The four cases (fixed by rules written before the source studies ran; re-resolved by the same rules after the reruns):

Case	Rule	Why it stresses the prediction	$\sigma_{\ln \beta}$	d_W	Class
D1	D1 optimum, $n_s = 2$ (s6 + s18, reference s6)	one transmissibility per bin	0.0627%	0.476	Identifiable
A1	case nearest the Identifiable/Marginal boundary	$\beta \times 2$ ring floe	0.0534%	0.445	Identifiable
R2	R2 stress sea ($\omega_p = 6$, $\gamma_J = 7$)	largest Welch bias of the R2 seas	0.0448%	0.558	Marginal
R1	p_0 at LSM6DSV16X noise	the measured-noise operating point	0.0499%	0.436	Identifiable

After the amendment every case is limited by bias rather than precision, so M1 now tests the bias prediction as much as the scatter. The R2 case is Marginal by design: it is the sea with the largest predicted Welch bias.

Method (pre-registered in `cfg.M1`; unchanged).

1. **Stage 1.** Each case is rebuilt by its source study’s own code (`resolveM1Scenarios`) and evaluated by the frozen engine. It must reproduce its source study before any record is generated. All four did, to a relative difference of at most 2×10^{-16} in σ .
2. **Stage 2.** 100 records per case from `synthesiseTwinRecords` (fixed seed schedule), corrected transmissibility features from G3’s estimator, and the linearised estimate $\Delta\theta = F^{-1}J^T\Sigma^{-1}(z - f_0)$ with the *analytic* J , Σ_k and F . Pass: empirical over predicted σ in $[0.85, 1.20]$ for both parameters.
3. **Stage 3.** $q = \Delta\theta^T F \Delta\theta$, centred on the truth (as G3a, so $E[q] = 2 + d_W^2$), against χ_2^2 . Pass: 95% coverage in $[0.88, 0.99]$. The mean offset against $\delta\theta_W$ is a diagnostic, not a criterion. If a statistic’s 1-standard-error interval contains a limit, records 101 to 200 are added once (the “borderline” rule).
4. **Stage 4.** Record 1 is inverted with G3’s safeguarded Gauss–Newton solver from $\beta \times 0.8$ and $\times 1.2$ (R true). Pass: converged, and $|\theta_{NL} - \theta_{lin}|/\sigma < 0.2$ per parameter (G3.1).

Reuse, not reimplement. The estimator, information matrix, objective, Jacobian and solver are G3’s own, copied by a program into `g3EstimatorCore.m`. The tester checks every function body against the G3 source, and checks that the nonlinear model on the EMM reproduces the engine’s features at the A1 truth (1.1×10^{-15}). The run refused to start with uncommitted code, so commit 1b61a0f is the code that ran (96 min).

2 Results

Case	n	emp/pred (± 1 SE)		coverage		mean q ($2 + d_W^2$)	nonlinear (both starts)	Verdict
		$\ln \beta$	$\ln R$	68%	95%			
D1 $n_s = 2$	100	1.011 ± 0.072	0.982 ± 0.070	0.620	0.940	2.17 (2.23)	conv., conv.	pass
A1 $\beta \times 2$	200*	1.062 ± 0.053	1.112 ± 0.056	0.620	0.915	2.35 (2.20)	conv., conv.	pass
R2 stress sea	200*	0.913 ± 0.046	0.929 ± 0.047	0.695	0.950	2.03 (2.31)	conv., <i>not flagged</i>	fail
R1 p_0 , LSM	100	0.940 ± 0.067	1.015 ± 0.072	0.630	0.920	2.33 (2.19)	conv., conv.	pass

*Extended by the borderline rule on emp/pred at 100 records. In every run the nonlinear end point equals the linearised estimate to better than 0.0005σ . The R2 sea loses two of 84 bins on average (82 accepted in the analytic target); 30 of its 200 records had a further rejected bin and were estimated without it, as registered. Nominal coverage 0.683 and 0.950.

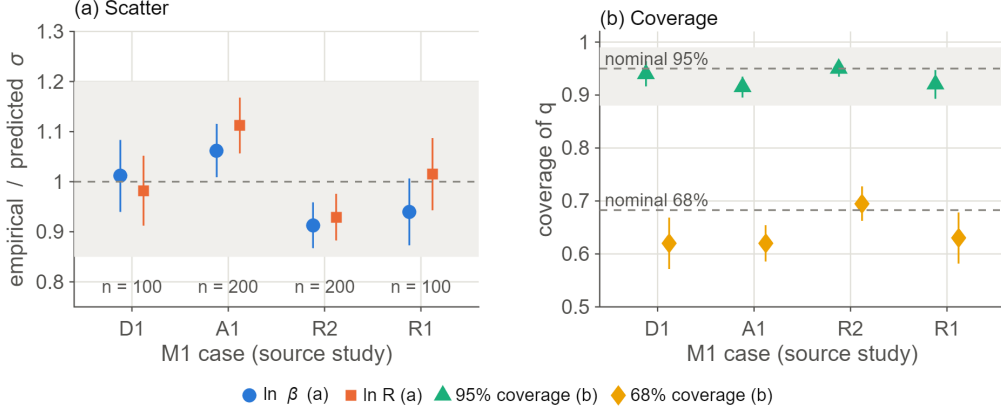


Figure 1. Acceptance per case. (a) Empirical over predicted standard deviation with ± 1 standard error; shaded: the pre-registered band. (b) Coverage of the predicted 68% and 95% regions with binomial ± 1 standard error; shaded: the 95% acceptance band.

Scatter and coverage. All eight scatter ratios lie inside the band, on both sides of 1. The largest departures are:

- A1 in $\ln R$: 1.112, 2.0 standard errors above 1;
- R2 in $\ln \beta$: 0.913, 1.9 standard errors below 1.

Both cases needed the extension to 200 records. All 95% coverages lie inside the band.

2.1 Welch-bias prediction

Mean offset of the linearised estimates against the Welch-expected bias (a diagnostic, not a criterion; G3.3 rule: within 3 standard errors).

Case	$\ln \beta$ (%)			$\ln R$ (%)		
	Welch-expected	empirical mean $\pm$ SE	difference	Welch-expected	empirical mean $\pm$ SE	difference
D1	+0.0235	$+0.0263 \pm 0.0063$	+0.4 SE	+0.0068	$+0.0079 \pm 0.0014$	+0.8 SE
A1	+0.0205	$+0.0243 \pm 0.0040$	+1.0 SE	+0.0040	$+0.0042 \pm 0.0007$	+0.3 SE
R2	+0.0186	$+0.0222 \pm 0.0029$	+1.2 SE	+0.0038	$+0.0042 \pm 0.0004$	+1.0 SE
R1	+0.0199	$+0.0292 \pm 0.0047$	+2.0 SE	+0.0035	$+0.0044 \pm 0.0008$	+1.1 SE

Every offset is consistent with $\delta\theta_W$ individually. However, all eight empirical means lie *above* the prediction. If they were independent and unbiased, the chance of eight of one sign would be about 1 in 128. The excess is 5 to 47% of the Welch bias, or 0.02 to 0.19σ .

This suggests that the analytic Welch bias is slightly underpredicted. A likely source is a finite-segment ratio bias that the smoothing-only prediction omits. The effect is small but not negligible now that classes are bias-limited: at p_0 it would raise d_W by about 0.1 to 0.2. M1 does not resolve it on its own. A larger Monte Carlo reference (the E1 covariance and bias references) is the right test.

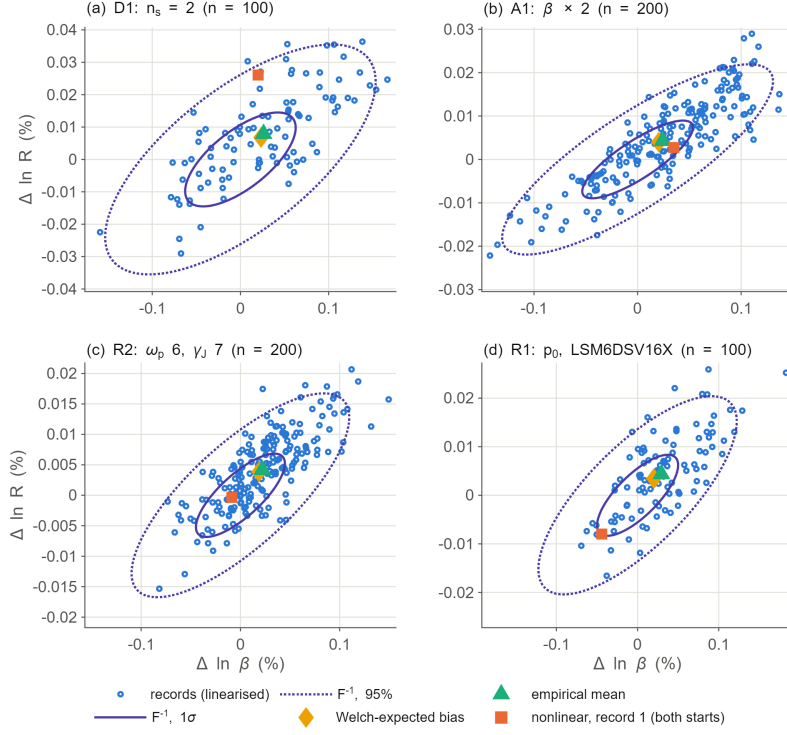


Figure 3. Per-record linearised estimates (minus the truth) against the analytic 1σ and 95% ellipses of F^{-1} , with the Welch-expected bias, the empirical mean and the nonlinear estimates of record 1 (both starts coincide). The strong β - R correlation predicted by F is reproduced in every case.

2.2 Nonlinear spot checks

The starts lie 290 to 500 predicted $\sigma_{\ln \beta}$ from the truth (20% in β against σ of 0.045 to 0.063%). In all eight runs the solver reached the linearised estimate of the same record (gap below 0.0005σ) in 4 or 5 iterations (18 to 29 model evaluations, 10 to 16 min each), with χ^2/dof 0.88 to 0.99. Both starts of each case end at the same point, within 1.9σ of the truth.

The failing run. The R2 case from $\beta \times 1.2$ ended after 5 iterations flagged “not converged”. Its end point, χ^2/dof and distance from the truth ($[-0.19, -0.05]\sigma$) are the same as those of the $\beta \times 0.8$ start, which converged. G3’s solver (used verbatim) sets the convergence flag only when an *accepted* step is below 0.01σ . It also stops when, with a freshly computed Jacobian, no step length (1, 0.5, 0.25, 0.125) lowers the objective, a branch its own code comments as “at the minimum”, but this exit leaves the flag false. The stored iteration history (`diagnoseM1Convergence`, no EMM solves) confirms this. In every other run, the third accepted step was 0.04 to 0.6σ , and a fourth step of 10^{-6} to $3 \times 10^{-5}\sigma$ still lowered the objective in the ninth decimal, which set the flag. In the failing run the third step was 0.25σ . The next step, presumably as small as in the other runs, did not lower the objective even with a fresh Jacobian, so the solver stopped. Its end point differs from the linearised estimate by $6 \times 10^{-5}\sigma$ and from the other start’s by $2 \times 10^{-4}\sigma$, and its objective (465.85232575) is marginally *lower* than that of the converged run (465.85232585). The run was at the minimum to floating-point precision. The estimate is correct; the label is not. Under the pre-registered rule (converged *and* within 0.2σ) the case fails, and is reported as such.

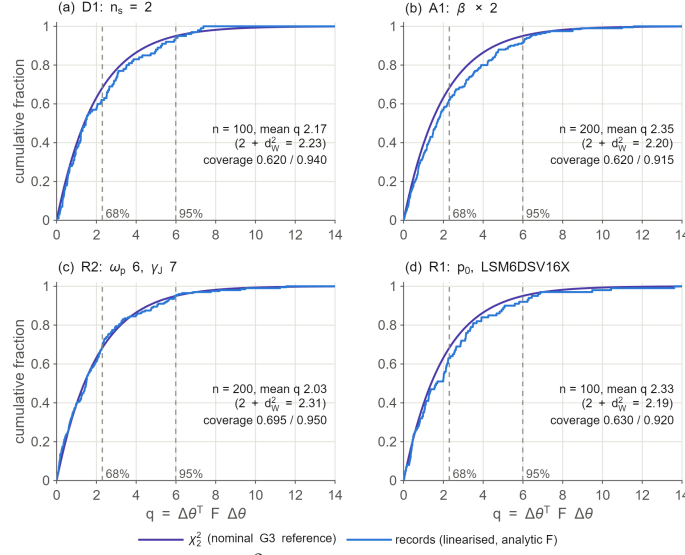


Figure 2. Empirical distribution of q per case against χ^2_2 , the nominal coverage reference of G3 (dashed: its 68% and 95% quantiles). Because q is centred on the truth and the estimator carries a Welch bias, q is not exactly central χ^2_2 : its expected mean is $2 + d_W^2$ (given per panel).

3 Interpretation

1. **The analytic uncertainty holds where it was most likely to fail.** The minimal two-sensor design, the A1 boundary floe, the most biased R2 sea and the measured-noise operating point all reproduce the predicted scatter, ellipse and coverage within sampling error. The scatter ratios fall on both sides of 1, with no consistent sign.
2. **The bias prediction holds case by case but may be slightly low.** All eight offsets are within 2 standard errors, but all lie above the prediction, by up to 0.19σ . Before the amendment this would have been irrelevant. Now that the classes are bias-limited, a 10 to 45% underprediction of $\delta\theta_W$ could move cases near $d_W = 0.5$. It should be quantified by the E1 Monte Carlo references before the thesis quotes d_W to two decimals.
3. **The inverse is well behaved within $\pm 20\%$ in β .** From hundreds of σ away, every inversion reached the linearised estimate. No second minimum was met, and the linear theory used everywhere else applies at the solution.
4. **The one failure is a solver-termination label, not an estimate.** The R2 run from $\beta \times 1.2$ ended at the same point as the converged run. A pre-registered rule that requires the flag is applied as written; the thesis should report the failure and its cause in one sentence.

Conclusion (thesis wording). M1 tested the analytic inverse outside its original p_0 validation regime, in four pre-registered stress cases: the minimum two-sensor design, the A1 boundary floe, the most strongly biased R2 spectrum and the measured LSM6DSV16X noise. Empirical scatter stayed within 11% of the Fisher prediction and the 95% coverage within the pre-registered interval in every case. Mean offsets agreed with the predicted Welch bias within two standard errors each, although all eight lay slightly above it, suggesting that the analytic bias may be underestimated by up to about 0.2σ . All eight nonlinear inversions reached the linearised estimate from $\pm 20\%$ stiffness starts. One of them, for the R2 case, terminated through the solver’s no-descent exit and was therefore not flagged as converged, so that case fails the registered criterion despite an identical estimate.

Limitations.

- Within-model: truth and inverse share the EMM, so M1 verifies the estimator and the Fisher/bias prediction, not the physics (V1).
- Four deliberately selected stress cases: a spot check, not an exhaustive replication of every scenario.
- 100 to 200 records resolve scatter ratios to about ± 5 to 7%, and bias offsets to about 0.06 to 0.13σ .
- The nonlinear check uses one record per case and two starts in β only.
- Records use the prediction’s node grid by design.

Requirement for the experiment and the thesis

Use the analytic Fisher predictions for the design, robustness and domain conclusions. Treat the analytic Welch bias as possibly underestimated by up to about 0.2σ , pending the E1 Monte Carlo references, and be cautious with classes that sit close to $d_W = 0.5$. The linearised and nonlinear estimates are interchangeable at the solution, and the inverse converges from $\pm 20\%$ in β . For real data, judge solver convergence on the step and the objective, not on the flag alone.

Output (Results/M1/)	Contents
M1_20261009_143730.mat / _summary.txt	results, config, every per-record estimate, nonlinear histories
M1_analytic.csv, M1_summary.csv	targets vs sources; acceptance statistics per case
M1_records.csv, M1_nonlinear.csv	per-record estimates; nonlinear runs
checkpoints/, figures/M1_fig1 to M1_fig3	resumable state; Figures 1 to 3 (plotM1MonteCarloSpotChecks)
Superseded	M1_20261003_221127 (pre-amendment)